Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: MACHADO & SONS

Physical address of dairy:

Number and Street	City	County	Zip Code
11735 CARROLL RD	ELK GROVE	Sacramento	95757

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 02/09/1972

Regional Water Quality Control Board Basin Plan designation: Sacramento River Basin

County Assessor Parcel Number(s) for dairy facility:

X132-X240-X061-XXXX

B. OPERATORS

Machado, David

Operator name: Machado, David

Telephone no.: Landline: ; Cellular: (916) 715-9102

Mailing Address Number and Street	City	State	Zip Code
11735 Carroll RD	Elk Grove	CA	95757

Machado, Dusty

Operator name: Machado, Dusty

Telephone no.: Landline: ; Cellular: (916) 997-2432

Mailing Address Number and Street	City	State	Zip Code
11735 Carroll RD	Elk Grove	CA	95757

This operator is responsible for paying permit fees.

C. OWNERS

Machado, David

Legal owner name: Machado, David

Telephone no.: Landline: ; Cellular: (916) 715-9102

Mailing Address Number and Street	City	State	Zip Code
11735 Carroll RD	Elk Grove	CA	95757

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	15	480	0	0
Number under roof	110	10	0	0	20	15
Maximum number	110	10	15	480	20	15
Average number	110	10	15	480	20	15
Avg live weight (lbs)	1,400	1,500	1,100	775		

Predominant milk cow breed: Holstein

Average milk production: 65 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 7,850.86 *tons per reporting period*

Total nitrogen from manure: 86,636.86 *lbs per reporting period*

After ammonia losses (30% loss applied): 60,645.80 *lbs per reporting period*

Total phosphorus from manure: 14,407.45 *lbs per reporting period*

Total potassium from manure: 20,704.66 *lbs per reporting period*

Total salt from manure: 54,093.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 4,416,000 *gallons*

Total nitrogen generated: 6,972.95 *lbs*

Total phosphorus generated: 2,279.59 *lbs*

Total potassium generated: 29,362.65 *lbs*

Total salt generated: 161,649.99 *lbs*

	4,416,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	4,416,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Dairy/Dom.	Ground water
Joey Ag Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field#1	49	49	4	both	X132-X240-X061-XXXX
Field#2	17	17	2	both	X132-X240-X061-XXXX
Field#3	55	55	0	none	X132-X240-X061-XXXX
Totals for areas that were used for application	66	66	6		
Totals for areas that were not used for application	55	55	0		
Land application area totals	121	121	6		

B. CROPS AND HARVESTS

Field#1

Field name: Field#1

10/31/2022: Pasture mix

Crop: Pasture mix Acres planted: 49 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	480.00 *ton*	Dry-weight		78.6	28,700.00	3,940.00	23,700.00		12.30
05/05/2023	108.00 *ton*	Dry-weight		78.6	27,000.00	4,020.00	23,400.00		11.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	10.00	116.00	17.00	100.00	0.00
Total actual harvest content	12.00	145.80	20.31	121.44	627.01

Field#1

06/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 49 Plant date: 06/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/26/2023	1,000.00 *ton*	Dry-weight		71.0	10,600.00	2,810.00	11,000.00		6.91
10/26/2023	225.00 *ton*	Dry-weight		71.6	10,500.00	2,510.00	9,820.00		6.55

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	43.20	172.80	0.00
Total actual harvest content	25.00	152.86	39.81	155.82	988.75

Field#2

Field name: Field#2

10/31/2022: Pasture mix

Crop: Pasture mix Acres planted: 17 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	204.00 *ton*	Dry-weight		76.6	27,000.00	4,020.00	23,400.00		11.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	10.00	123.00	17.00	100.00	0.00
Total actual harvest content	12.00	151.63	22.58	131.41	662.69

06/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 06/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/26/2023	425.00 *ton*	Dry-weight		71.6	10,500.00	2,510.00	9,820.00		6.55

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	43.20	172.80	0.00
Total actual harvest content	25.00	149.10	35.64	139.44	930.10

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field#1 - 10/31/2022: Pasture mix

Field name: Field#1

Crop: Pasture mix — Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Separator solids	156.86	77.50	346.58	1,884.80	490.00 *ton*
Application event totals		156.86	77.50	346.58	1,884.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/13/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	11.05	2.60	34.22	157.28	624,000.00 *gal*
Application event totals		11.05	2.60	34.22	157.28	

Field#1 - 06/22/2023: Corn, silage

Field name: Field#1

Crop: Corn, silage — Plant date: 06/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	91.82	31.29	402.13	2,252.95	2,496,000.00 *gal*
Joey Ag Well	Ground water	0.06	0.00	0.00	0.00	2,500,000.00 *gal*
Application event totals		91.88	31.29	402.13	2,252.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Joey Ag Well	Ground water	0.14	0.00	0.00	0.00	5,400,000.00 *gal*
Application event totals		0.14	0.00	0.00	0.00	

Field#1 - 06/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Joey Ag Well	Ground water	0.13	0.00	0.00	0.00	5,010,000.00 *gal*
Application event totals		0.13	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
COMMERCIAL FERTILIZER SIDEDRESS	Solid commercial fertilizer	75.00	0.00	0.00	0.00	
Joey Ag Well	Ground water	0.13	0.00	0.00	0.00	5,010,000.00 *gal*
Application event totals		75.13	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Joey Ag Well	Ground water	0.13	0.00	0.00	0.00	5,010,000.00 *gal*
Application event totals		0.13	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Joey Ag Well	Ground water	0.13	0.00	0.00	0.00	5,010,000.00 *gal*
Application event totals		0.13	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Joey Ag Well	Ground water	0.13	0.00	0.00	0.00	5,010,000.00 *gal*
Application event totals		0.13	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Joey Ag Well	Ground water	0.13	0.00	0.00	0.00	5,010,000.00 *gal*
Application event totals		0.13	0.00	0.00	0.00	

Field#2 - 10/31/2022: Pasture mix

Field name: Field#2

Crop: Pasture mix

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Separator solids	156.86	77.50	346.58	1,884.80	170.00 *ton*
Application event totals		156.86	77.50	346.58	1,884.80	

02/14/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	22.05	5.20	68.28	313.85	432,000.00 *gal*
Application event totals		22.05	5.20	68.28	313.85	

Field#2 - 06/22/2023: Corn, silage

Field name: Field#2

Crop: Corn, silage

Plant date: 06/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	91.61	31.22	401.22	2,247.85	864,000.00 *gal*
Joey Ag Well	Ground water	0.15	0.00	0.00	0.00	2,100,000.00 *gal*
Application event totals		91.77	31.22	401.22	2,247.85	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Joey Ag Well	Ground water	0.16	0.00	0.00	0.00	2,220,000.00 *gal*
Application event totals		0.16	0.00	0.00	0.00	

Field#2 - 06/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Joey Ag Well	Ground water	0.13	0.00	0.00	0.00	1,800,000.00 *gal*
Application event totals		0.13	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
COMMERCIAL FERTILIZER SIDEDRESS	Solid commercial fertilizer	75.00	0.00	0.00	0.00	
Joey Ag Well	Ground water	0.13	0.00	0.00	0.00	1,800,000.00 *gal*
Application event totals		75.13	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Joey Ag Well	Ground water	0.13	0.00	0.00	0.00	1,800,000.00 *gal*
Application event totals		0.13	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Joey Ag Well	Ground water	0.13	0.00	0.00	0.00	1,800,000.00 *gal*
Application event totals		0.13	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Joey Ag Well	Ground water	0.13	0.00	0.00	0.00	1,800,000.00 *gal*
Application event totals		0.13	0.00	0.00	0.00	

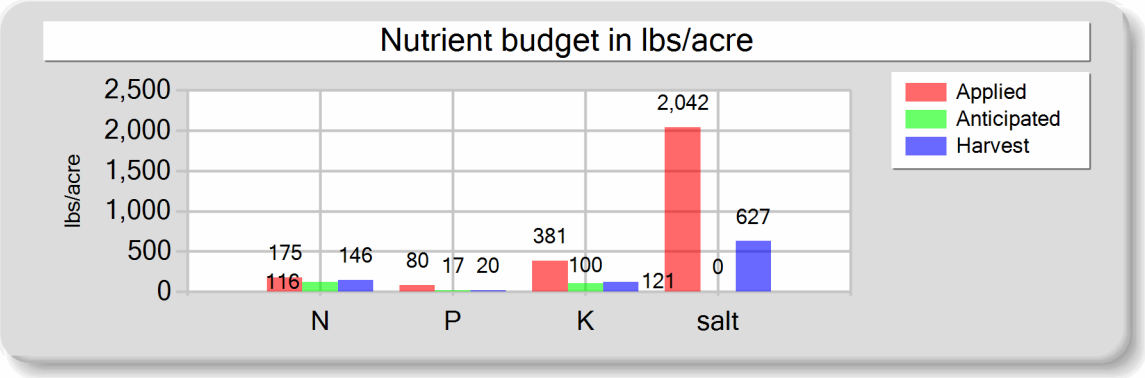
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Joey Ag Well	Ground water	0.13	0.00	0.00	0.00	1,800,000.00 *gal*
Application event totals		0.13	0.00	0.00	0.00	

B. NUTRIENT BUDGET

Field#1 - 10/31/2022: Pasture mix

Field name: Field#1 Crop: Pasture mix Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	156.86	77.50	346.58	1,884.80
Process wastewater	11.05	2.60	34.22	157.28
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	174.91	80.10	380.80	2,042.08
Anticipated crop nutrient removal	116.00	17.00	100.00	0.00
Actual crop nutrient removal	145.80	20.31	121.44	627.01
Nutrient balance	29.11	59.79	259.36	1,415.07
Applied to removed ratio	1.20	3.94	3.14	3.26

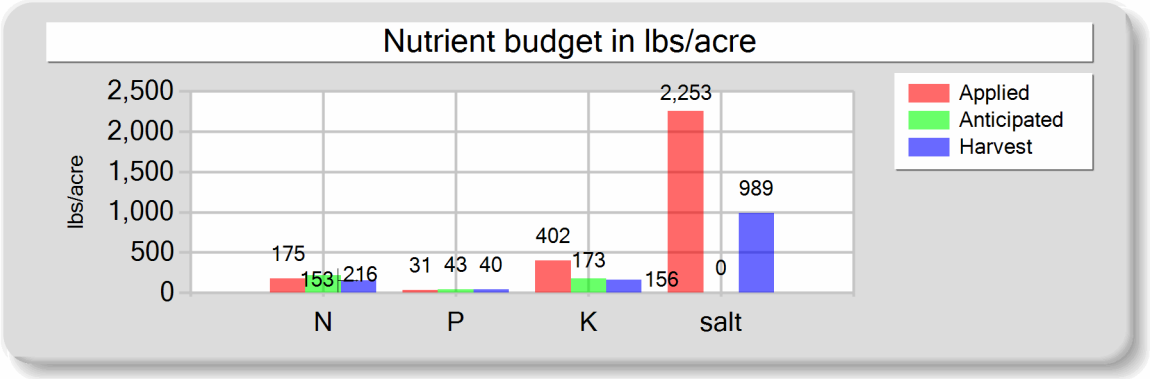
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
624,000.00 *gallons*
22.98 *acre-inches*
0.47 *inches/acre*

Total harvests for the crop
2 *harvests*

Field#1 - 06/22/2023: Corn, silage

Field name: Field#1 Crop: Corn, silage Plant date: 06/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	75.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	91.82	31.29	402.13	2,252.95
Fresh water	0.97	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	174.79	31.29	402.13	2,252.95
Anticipated crop nutrient removal	216.00	43.20	172.80	0.00
Actual crop nutrient removal	152.86	39.81	155.82	988.75
Nutrient balance	21.93	-8.52	246.31	1,264.19
Applied to removed ratio	1.14	0.79	2.58	2.28

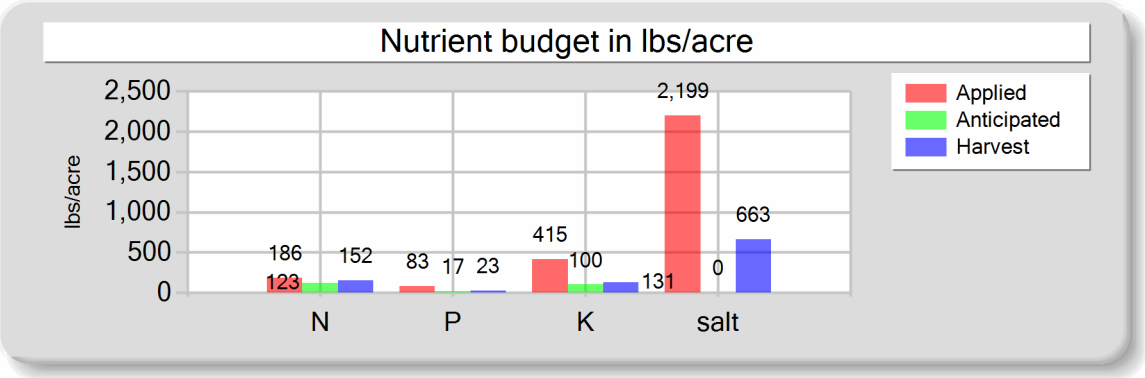
Fresh water applied	
37,960,000.00	*gallons*
1,397.94	*acre-inches*
28.53	*inches/acre*

Process wastewater applied	
2,496,000.00	*gallons*
91.92	*acre-inches*
1.88	*inches/acre*

Total harvests for the crop	
2	*harvests*

Field#2 - 10/31/2022: Pasture mix

Field name: Field#2 Crop: Pasture mix Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	156.86	77.50	346.58	1,884.80
Process wastewater	22.05	5.20	68.28	313.85
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	185.91	82.70	414.86	2,198.65
Anticipated crop nutrient removal	123.00	17.00	100.00	0.00
Actual crop nutrient removal	151.63	22.58	131.41	662.69
Nutrient balance	34.28	60.12	283.45	1,535.96
Applied to removed ratio	1.23	3.66	3.16	3.32

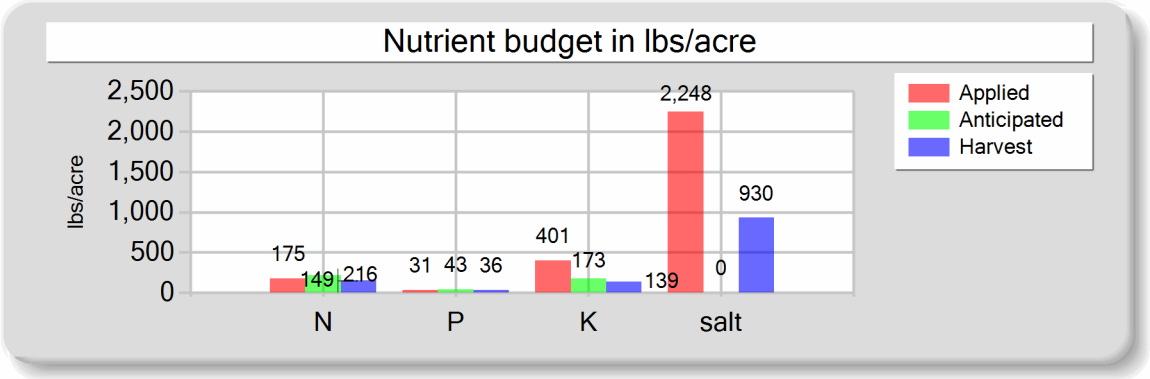
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
432,000.00	*gallons*
15.91	*acre-inches*
0.94	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field#2 - 06/22/2023: Corn, silage

Field name: Field#2 Crop: Corn, silage Plant date: 06/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	75.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	91.61	31.22	401.22	2,247.85
Fresh water	1.11	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	174.72	31.22	401.22	2,247.85
Anticipated crop nutrient removal	216.00	43.20	172.80	0.00
Actual crop nutrient removal	149.10	35.64	139.44	930.10
Nutrient balance	25.62	-4.43	261.78	1,317.75
Applied to removed ratio	1.17	0.88	2.88	2.42

Fresh water applied	
15,120,000.00	*gallons*
556.82	*acre-inches*
32.75	*inches/acre*

Process wastewater applied	
864,000.00	*gallons*
31.82	*acre-inches*
1.87	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

T2348024B

Sample and source description: T2348024B

Sample date: 12/14/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,300.00	12,500.00	55,900.00	23,800.00	13,900.00	5,300.00	8,200.00	1,660.00		30.40
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

T3130030B

Sample and source description: T3130030B

Sample date: 05/10/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 19.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,640.00	3,600.00	6,070.00							0.00
DL	1.00	1.00	1.00							1.00

T3307007C

Sample and source description: T3307007C

Sample date: 11/03/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 40.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,100.00	20,100.00	39,900.00							0.00
DL	1.00	1.00	1.00							1.00

B. PROCESS WASTEWATER ANALYSES

T3059014A

Sample and source description: T3059014A

Sample date: 02/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	104.00	28.00	0.00	0.00	24.50	322.00								1,500.00	1,480
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3130030A

Sample and source description: T3130030A

Sample date: 05/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	216.00	78.40	0.00	0.00	73.60	946.00								5,090.00	5,300
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

C. FRESH WATER ANALYSES

Dairy/Dom.

T3314019B

Sample description: T3314019B

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	8.02	0.00	8.02								771.00	0
DL	1.00	0.50	1.00								100.00	1

Joey Ag Well

Joey Ag Well

T3314019A

Sample description: T3314019A

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.15	0.00	0.15								232.00	0
DL	0.10	0.50	0.10								100.00	1

D. SOIL ANALYSES

Field#1

T3349021A

Sample and source description: T3349021A

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			66.00		0.00	0.00	
DL			1.00		1.00	1.00	

T3349021B

Sample and source description: T3349021B

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			82.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field#2

Field#2

T3349021B

Sample and source description: T3349021B

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			82.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field#3

T3349021B

Sample and source description: T3349021B

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			82.00		0.00	0.00	
DL			1.00		1.00	1.00	

T3349021C

Sample and source description: T3349021C

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			73.00		0.00	0.00	
DL			1.00		1.00	1.00	

E. PLANT TISSUE ANALYSES

Field#1 - 10/31/2022: Pasture mix

Field#1 - 10/31/2022: Pasture mix

T3130030C

Sample and source description: T3130030C

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,000.00	4,020.00	23,400.00		11.80
DL	1.00	1.00	1.00		1.00

T3130030D

Sample and source description: T3130030D

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,700.00	3,940.00	23,700.00		12.30
DL	1.00	1.00	1.00		1.00

Field#1 - 06/22/2023: Corn, silage

T3307007A

Sample and source description: T3307007A

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,600.00	2,810.00	11,000.00		6.91
DL	1.00	1.00	1.00		1.00

Field#1 - 06/22/2023: Corn, silage

T3307007B

Sample and source description: T3307007B

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,500.00	2,510.00	9,820.00		6.55
DL	1.00	1.00	1.00		1.00

Field#2 - 10/31/2022: Pasture mix

T3130030C

Sample and source description: T3130030C

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,000.00	4,020.00	23,400.00		11.80
DL	1.00	1.00	1.00		1.00

Field#2 - 06/22/2023: Corn, silage

T3307007B

Sample and source description: T3307007B

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,500.00	2,510.00	9,820.00		6.55
DL	1.00	1.00	1.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

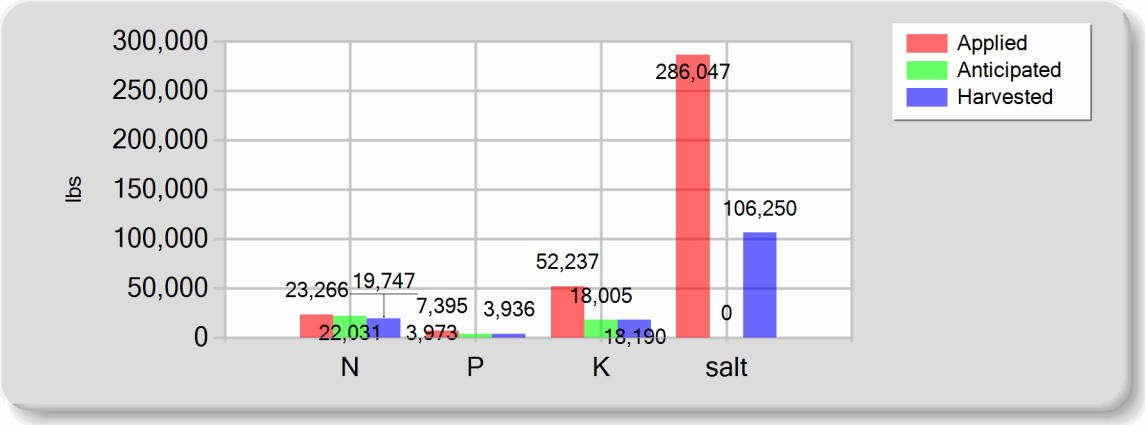
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

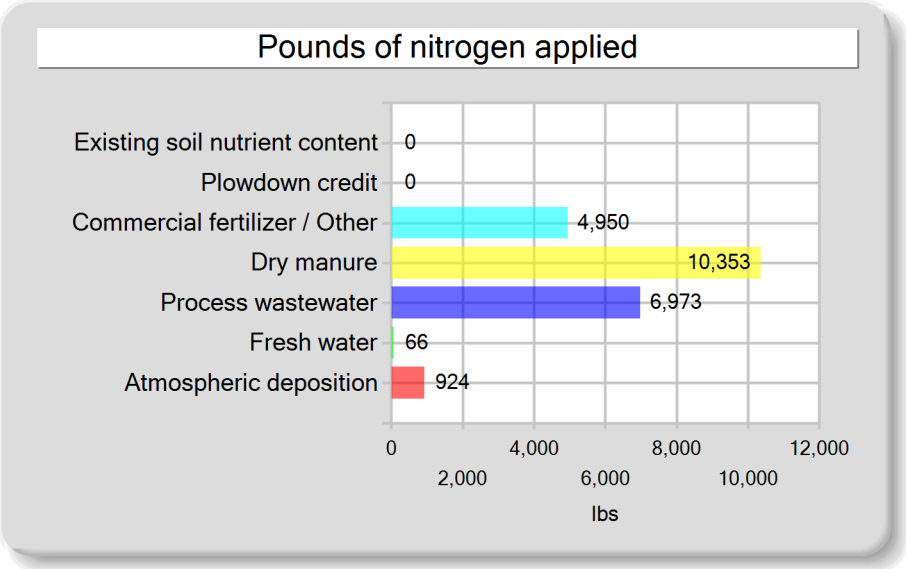
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

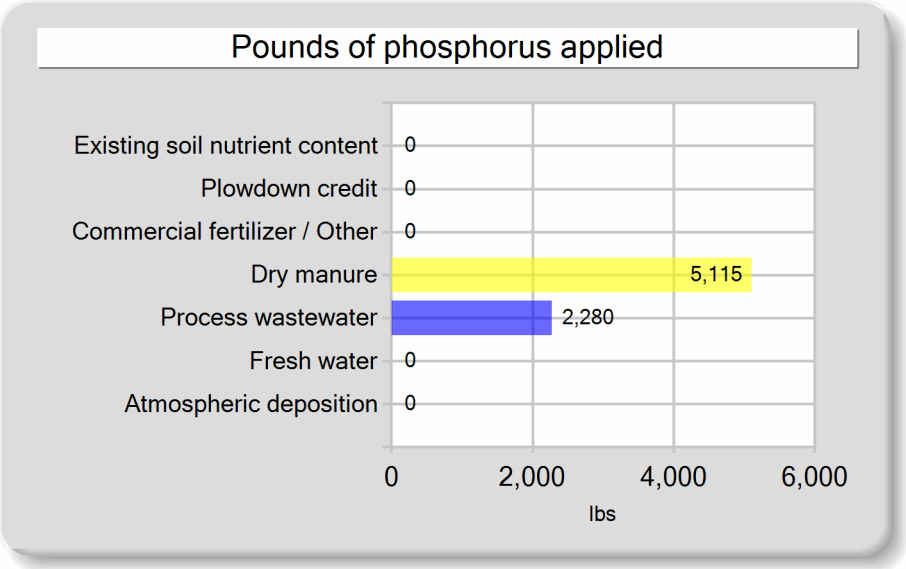
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	4,950.00	0.00	0.00	0.00
Dry manure	10,352.76	5,115.00	22,874.28	124,396.80
Process wastewater	6,972.95	2,279.59	29,362.65	161,649.99
Fresh water	66.44	0.00	0.00	0.00
Atmospheric deposition	924.00	0.00	0.00	0.00
Total nutrients applied	23,266.15	7,394.59	52,236.93	286,046.79
Anticipated crop nutrient removal	22,031.00	3,973.20	18,004.80	0.00
Actual crop nutrient removal	19,746.52	3,935.54	18,190.16	106,249.85
Nutrient balance	3,519.63	3,459.04	34,046.77	179,796.95
Applied to removed ratio	1.18	1.88	2.87	2.69

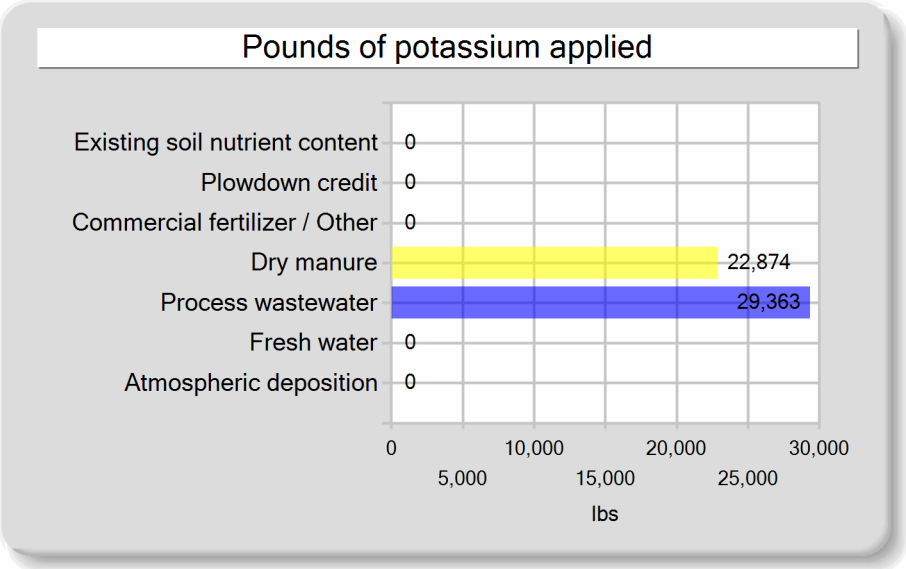
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

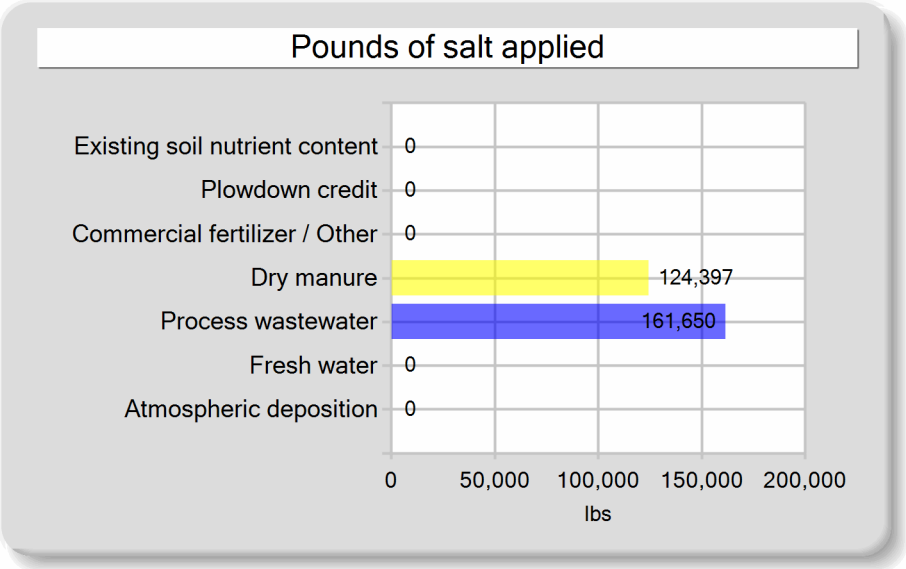


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY

Dusty Machado
PRINT OR TYPE NAME

6/23/24
DATE

Same as Owner
SIGNATURE OF OPERATOR OF FACILITY

Dusty Machado
PRINT OR TYPE NAME

DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: MACHADO AND SONS DAIRY
Mailing address: 11735 CARROLL RD.
(not facility address) ELK GROVE, CA 95757
Phone#: (916) 715-9102

Authorized Copy Released To:
JCRK, INC.
(209) 327-0992
Sampled By: Dusty Machado
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	11/9/23	1:00p	Joey Ag Well	Ag Well	11.6°C	20 min	X				
2	✓	1:30p	Dairy/Domestic	Dom. Well	11.0	✓	X				
3											
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	Jacquie Captein	JCRK		11/10/23 8:00a		
2nd		DENELE ANALYTICAL, INC.	11-10-23 1:00			T3314019

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Machado & Sons Dairy-Elk Grove
11735 Carroll Rd
Elk Grove, CA 95757

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/14/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 11.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Joey Ag Well	T3314019A	Electrical Conductivity (EC)	EPA 120.1 and	0.232	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.151	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	148	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.6			
Electrical Conductivity (EC)	93.2			1.39
Nitrate Nitrogen (NO3-N)	102	100 (T3314015D)	100.0 (T3314015D)	0.459

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Machado & Sons Dairy-Elk Grove
11735 Carroll Rd
Elk Grove, CA 95757

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/14/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dairy/Dom.	T3314019B	Electrical Conductivity (EC)	EPA 120.1 and	0.771	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	8.02	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	494	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.6			
Electrical Conductivity (EC)	93.2			1.39
Nitrate Nitrogen (NO3-N)	102	100 (T3314015D)	100.0 (T3314015D)	0.459

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